



An effective heuristic for flexible job-shop scheduling problem with maintenance activities [☆]

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ABSTRACT

Most production scheduling problems, including the standard flexible job-shop scheduling problem (FJSP), assume that machines are continuously available. However, in most realistic situations, machines may become unavailable during certain periods due to preventive maintenance (PM). In this paper, a flexible job-shop scheduling problem with machine availability constraints is considered. Each machine is subject to preventive maintenance during the planning period and the starting times of maintenance activities are either flexible in a time window or fixed beforehand. Moreover, two cases of maintenance resource constraint are considered: sufficient maintenance resource available or only one maintenance resource available. To deal with this variant FJSP problem with maintenance activities, a filtered beam search (FBS) based heuristic algorithm is proposed. With a modified branching scheme, the machine availability constraint and maintenance resource constraint can be easily incorporated into the proposed algorithm. Simulation experiments are conducted on some representative problems. The results demonstrate that the proposed filtered beam search based heuristic algorithm is a viable and effective approach for the FJSP with maintenance activities.

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1. Introduction

Due to their importance both in the fields of manufacturing industries and operations research, production scheduling and maintenance planning have been received considerable attention both in academia and in industry. However, in practice, production scheduling and maintenance planning are typically made independently despite the implicit common goal of maximizing equipment productivity between them (Cassady & Kutanoglu, 2003, 2005).

Production scheduling is to allocate a limited set of resources to a number of jobs with optimization of one or multiple system criteria over time, while taking various constraints into account. Most of them assume that machines are always available at constant speed. They either ignore equipment failure or treat it as a random event (Cassady & Kutanoglu, 2003). Although reactive scheduling (or rescheduling) considers response scheduling after machine breakdown, it does not consider the practical requirement of high reliability for some equipment (e.g., furnace in steel plant, thermal generating units in power system (Mohanta, Sadhu, & Chakrabarti, 2007)). Reactive scheduling does not radically improve the reliability of machines either. Moreover, too often rescheduling may make

the production system into a kind of chaos, because rescheduling leads to rearrangement of supporting resources (e.g., personnel, spare parts, and material handling devices). Therefore, a more realistic scheduling model should consider the machine availability constraint. This helps reducing the frequency of rescheduling and improving the robustness. Maintenance activities, especially preventive maintenance (PM), can restore the reliability of machines before it goes to breakdown. However, on one hand, maintenance activities take time that could otherwise be used for production scheduling; on the other hand, industrial applications cannot ignore the forecasted maintenance operations on the machines (Mauguiere, Billaut, & Bouquard, 2005). Therefore, the problem of simultaneously scheduling production jobs and maintenance activities obtains great attentions in recent years.

Preventive maintenance planning is usually generated in a maintenance system, like Computerized Maintenance Management Systems (CMMS) and Asset Management System (AMS). Preventive maintenance planning requires some pre-scheduled or reserved unavailability interval of machines. Therefore, production scheduling with maintenance activities is also treated as availability constraint of machines in the literature. Schmidt (2000) has reviewed most results related to deterministic scheduling problems with availability constraints published before 1998. After Schmidt's work, many efforts are devoted to the production scheduling with maintenance activities in different machine environ-

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ments with different job characteristics and different optimality criteria.

In the single machine context, Graves and Lee (1999) considered the scheduling problem with *semiresumable*¹ jobs and two maintenance intervals of a fixed duration on a single machine to minimize either total weighted completion time or the maximum lateness. Cassady and Kutanoglu (2003) proposed an integrated model that simultaneously determines production scheduling and preventive maintenance for a single machine in terms of the total weighted tardiness of jobs. This model is further extended into a similar model in terms of total weighted completion time in Cassady and Kutanoglu (2005). Compared with the total enumeration procedure in Cassady and Kutanoglu (2005), Sortrakul, Nachtmann, and Cassady (2005) presented heuristics based on the genetic algorithm (GA) for the same model. The job characteristics considered in these three papers are *resumable* and non-preemptive for PM. Chen and Liao (2005) studied the deterministic and probabilistic models of a single machine problem with different maintenance situations and *resumable* jobs with respect to the minimization of number of tardy jobs. The research is employed into a textile company. Ji, He, and Cheng (2007) analyzed the worst-case ratio of the LPT (longest processing time) dispatching rule for a single machine scheduling problem with *nonresumable* jobs and several periodic maintenance in terms of makespan minimization. Chen (2008) presented two optimization models and a heuristic for a single machine scheduling problem with nonresumable jobs and with flexible and periodic maintenance to minimize the makespan. Chen (2009) proposed a heuristic based on Moore's algorithm and a branch-and-bound (B&B) algorithm to a single machine problem with periodic maintenance and *nonresumable* jobs with respect to the minimization of number of tardy jobs.

In the parallel machine context, Lee and Chen (2000) extended the work of Graves and Lee (1999) to parallel machines with *nonresumable* jobs to minimize total weighted completion time. Gharbi and Haouari (2005) considered the identical parallel-machine scheduling problem with availability restrictions for both jobs (release time and delivery time constraints) and machines. The objective is to minimize the makespan and they presented an exact branch-and-bound algorithm to solve this problem. With the objective of makespan minimization, Liao, Shyur, and Lin (2005) studied the two-parallel machine problem with only one machine unavailable for a fixed and known period. Both nonresumable and resumable cases are considered and the problem is solved with modified methods based on the Two-Machine Optimization (TMO) algorithm. Then, Lin and Liao (2007) extended the same problem to the case where each machine has a fixed and known unavailable period. Lee and Yu (2008) discussed a parallel-machine scheduling problem with the objectives of minimization of the expected sum of weighted completion times and the expected number of tardiness jobs, respectively. Both resumable and nonresumable cases are studied. Liao and Sheen (2008) analyzed the parallel-machine scheduling problem with machine availability and eligibility constraints while minimizing the makespan. The problem is formulated into a maximum flow problem and a binary search algorithm is proposed. Mellouli, Sadfi, Chu, and Kacem (2009) considered the identical parallel-machine scheduling problem with availability constraints and total completion time criterion. Three exact approaches (mixed integer linear programming based methods, a dynamic programming method and a

branch-and-bound method) are proposed as well as some heuristics adapted from the Shortest Processing Time (SPT) rule.

Because of their fundamental characteristic and simplicity, single machine and parallel-machine scheduling problems with availability constraints under different criteria have been extensively studied. Readers are referred to Ma, Chu, and Zuo (2010) for a relative comprehensive overview.

Recently, scheduling problems with availability constraints in more complicated environments, such as flow shop, job-shop and hybrid environments, have being attracted more and more attentions. Allaoui and Artiba (2004) studied the scheduling problem in a hybrid flow shop with both *resumable* and *nonresumable* jobs and maintenance constraints to minimize flow time and due date based criteria. Setup, cleaning and transportation times are also considered. Three dispatching rules, a simulated annealing (SA) heuristic and a flexible simulation model are separately used to solve the problem. Aggoune (2004) addressed a flow shop with machine availability constraints to minimize the makespan. Two variants of the non-preemptive constraints are considered: the starting times of maintenance activities are either fixed or flexible. An algorithm based on GA and tabu search (TS) is applied to solve the problem. Mauguier et al. (2005) presented a new job-shop scheduling problem with machine availability constraints. The jobs can be either *resumable* or *nonresumable* depending on the unavailability period. Kubzin and Strusevich (2006) studied both a two-machine open shop problem and a two-machine flow shop problem to minimize the makespan. Each machine is subject to preventive maintenance, and the length of each maintenance period depends on its starting time. Breit (2006) investigated a problem of scheduling n preemptive jobs in a two-machine flow shop where the first machine is not available for processing during a given time interval, with the minimization of makespan. Gao, Gen, and Sun (2006) considered a flexible job-shop scheduling problem (FJSP) with machine availability constraints. The periods of unavailability are non-fixed and flexible within an end-time window and have to be determined during the scheduling procedure. A hybrid genetic algorithm is proposed to solve the problem. Ruiz, Garcia-Diaz, and Maroto (2007) considered preventive maintenance and production scheduling in regular flow shops and six meta-heuristics are proposed with comparisons. The multi-purpose machine (MPM) scheduling problem with fixed machine availability constraints in a job-shop is studied in Zribi, Kamel, and Borne (2008). A two-phase heuristic method is proposed to solve the problem. In the first phase, a heuristic based on priority rules to solve the assignment problem is developed, which can be improved with a tabu search algorithm. In the second phase, a GA is presented to solve the sequencing problem. Yang, Hsu, and Kuo (2008) investigated a two-machine flow shop where maintenance activities with a constant time must be done after completing a fixed number of jobs at most. The objective is to find the optimal job combinations and the optimal job schedule to minimize the makespan. Naderi, Zandieh, and Ghomi (2009) investigated a job-shop scheduling problem with sequence-dependent set-up times and maintenance activities to minimize the makespan. Two meta-heuristics, simulated annealing (SA) and GA are used to solve the problem.

Based on the previous works review mentioned above, two gains could be drawn. Firstly, most existing literature focus on the problem of integrating production scheduling with preventive maintenance activities in the context of single machine, parallel machine and flow shop (especially two-machine problems). There is much less literature in the flexible job-shop manufacturing environment. As mentioned above, Gao et al. (2006) studied the general flexible job-shop scheduling problem combined with maintenance and production. Zribi et al. (2008) considered the MPM job-shop scheduling problem with availability constraints,

¹ *Resumable* means that if a job cannot be finished before a maintenance activity, it can continue the processing once the maintenance activity is finished. *Nonresumable* means a job must be reprocessed fully after the maintenance if its processing is interrupted by the maintenance activity on a machine. *Semiresumable* means that if a job is not processed to completion before the machine is stopped for maintenance, an additional set-up time is necessary when the processing is resumed.

which is the same problem as that of Gao et al. (2006). The hybrid flow shop with availability constraints in Allaoui and Artiba (2004) is one special type of FJSP with maintenance activities. Chan, Chung, Chan, Finke, and Tiwari (2006) considered distributed flexible manufacturing system (FMS) scheduling problems subject to machine maintenance constraint, in which the maintenance time is related to the machine age. This also covers the FJSP with maintenance activities. However, as compared with the models of single machine, parallel machine and flow shop, studies dealing with the FJSP with maintenance activities are rather rare.

The flexible job-shop extends the classical job-shop by overcoming the assumption of predetermining machine route and extends the classical parallel machines with multiple processing stages, to meet the real-world scenarios of production flexibility in a wide variety of products and faster production rates. The FJSP allows that a machine may be capable of performing more than one type of operation (Najib, Dauzere-Peres, & Zaidat, 2002). The FJSP at least reflects two practical scenarios such as flexible manufacturing systems (FMS) with multi-processor machines and scheduling of crane in port container terminals. The FJSP possesses two difficulties: how to assign each operation to one machine that is able to perform it (routing sub-problem) and how to determine starting times for each operation (sequencing sub-problem). Due to its significant theoretical and industrial importance, the FJSP has been explored to some extent in the literature since seminal work of Brandimarte (1993), such as Kacem, Hammadi, and Borne (2002a, 2002b), Xia and Wu (2005), Pezzella, Morganti, and Ciaschetti (2008), Gao, Sun, and Gen (2008), Wang, Zhou, and Xi (2008). Combined with the scheduling of maintenance activities and the standard FJSP provides additional complexity and new problems (Gao et al., 2006). More importantly, the integration scheduling of maintenance activities in the context of flexible job-shop environment is much more practical. Hence, it deserves further exploration to extend its theoretical significance and to explore its managerial implication.

Secondly, most existing literature consider the fixed machine availability constraint, that is, the starting time and the end time of maintenance activities are pre-determined in maintenance planning. However, in practice, preventive maintenance tasks would impose non-fixed availability constraints on the machine in that the starting time of these maintenance tasks is generally flexible (Gao et al., 2006). This means that the starting time of the unavailable periods is not known in advance and has to be determined within the given time window during the scheduling process. Currently, there is only a few works considered the non-fixed availability constraints, such as Graves and Lee (1999), Aggoune (2004), Kubzin and Strusevich (2006), Gao et al. (2006). Therefore, there is much room for further investigation.

Moreover, the constraints of maintenance resources (e.g., maintenance personnel, spare parts) should be considered. In a single machine scheduling problem with maintenance tasks, there is no need to consider this constraint since there is no conflict between maintenance resources and maintenance requirement. However, the conflict exists in the environment with multiple machines, and when only one machine can be maintained at any given time due to the limitation of maintenance resources. Lee and Chen (2000) considered the constraints of maintenance resources in parallel-machines scheduling with maintenance activities. In the FJSP with maintenance tasks discussed in Gao et al. (2006), the maintenance resource is assumed to be sufficient and the case of limited maintenance resource is not considered.

In this context, inspired by the work of Gao et al. (2006), and based on our previous work in FJSP (Wang et al., 2008), this paper proposes a heuristic based on the filtered beam search (FBS) algorithm to solve the flexible job-shop scheduling problem (FJSP) with the non-fixed and fixed machine availability constraints due to the

preventive maintenance, while considering two cases of maintenance resources constraints: sufficient maintenance resources or only one maintenance resource. In the first case, sufficient maintenance resources are available so that more than one machine can be maintained simultaneously if necessary. In the second case, only one machine can be maintained at any given time. A modified FBS procedure is presented to easily incorporate the machine availability constraint and the maintenance resource constraint.

The rest of the paper is organized as follows: Section 2 presents the formal description of the flexible job-shop scheduling problem with maintenance activities. Section 3 illustrates the specific FBS procedure for the problem under investigation. In Section 4, benchmarking examples are investigated with the proposed algorithm and comparisons are presented. In Section 5, conclusions are drawn and future work directions are given.

2. Statement of flexible job-shop scheduling problem with maintenance activities

The flexible job-shop scheduling problem with maintenance activities that we addressed here belongs to the broad field of the deterministic scheduling with machine availability constraints, and it can be normally described as follows.

A set of n jobs is to be processed on a set of m machines denoted by $M(M = \{M_1, M_2, \dots, M_m\})$. Each job j ($1 \leq j \leq n$) consists of a sequence of n_j operations (routing). Each operation O_{ij} ($1 \leq j \leq n, 1 \leq i \leq n_j$) has to be performed to complete one job. The process of each operation requires one machine out of a set M_{ij} of given machines, $M_{ij} \subseteq M$, and the processing time of an operation O_{ij} on the machine M_k is p_{ijk} ($M_k \in M_{ij} \subseteq M$). There are L_k ($k = 1, 2, \dots, m$) maintenance activities PM_u^k ($u \in L_k$) that have to be processed on each machine M_k during the planning horizon based on a certain maintenance policy (e.g., periodical policy, sequential policy (Wang, 2002) or preventive maintenance policy in Ruiz et al. (2007)). In the case of non-fixed availability constraint, the starting time of each maintenance task is within a predefined time window $[\underline{S}_u^k, \bar{S}_u^k]$ ($k = 1, 2, \dots, m, u \in L_k$) ($\underline{S}_u^k < \bar{S}_u^k$), and the duration of the maintenance task is pre-determined as pm_u^k . While in the case of fixed availability constraint, $\underline{S}_u^k = \bar{S}_u^k$. The processing is *nonresumable*, i.e. a job must be reprocessed fully after the maintenance if its processing is interrupted by the maintenance activity on the machine.

The FJSP with maintenance activities is to determine an assignment of the operations on the machines, a sequence of the operations, and maintenance tasks on the machines that optimizes one or several criteria. In this study, the following three criteria are to be minimized:

- F_1 : Makespan or maximal completion time of machines.
- F_2 : Total workload of the machines, which represents the total working time of all machines.
- F_3 : Critical machine workload, i.e. the workload of the most loaded machine.

These three criteria were considered in literature on the standard FJSP, like Kacem et al. (2002b), Xia and Wu (2005), Wang et al. (2008). These criteria are conflicting, which is explained in our previous work of Wang et al. (2008). In this work, the maintenance periods are also included in these three criteria as Gao et al. (2006) did, because the maintenance periods are mandatory and the decision maker has to schedule the maintenance periods along with the jobs to be processed.

Assumptions considered in this paper are given in the following:

1. The jobs are non-preemptive.

- The jobs are independent of each other and all jobs can be performed at r_j (release time).
- All machines are independent of each other and are available at $t = 0$.
- An operation cannot be performed by more than one machine at the same time and each machine can perform only one operation at a time.
- The set-up and removal times are sequence independent and included in the processing times or can be neglected.
- The transportation times of operations among machines are neglected.
- There is an unlimited in-progress product inventory, in which a job can wait for a machine until it is available.
- There are two cases of maintenance resources constraint. One case is that there are sufficient maintenance resources such that any number of machines can be maintained or repaired simultaneously. Another case is that the maintenance resources are limited so that only one machine can be maintained at any given time and hence the maintenance activity on one machine cannot be overlapped with that of another machine.
- When a maintenance task is performed on a machine, no operation can be processed on that machine.
- In the planning horizon, there is at least one maintenance task on a machine. The earliest starting time and the latest starting time satisfy $\underline{S}_u^k \leq \bar{S}_u^k$.

3. The proposed heuristic method

In the following subsection, the basic idea of filtered beam search is recalled from Wang et al. (2008). The readers familiar with this model can go directly to Section 3.2 on the modified FBS-based heuristic for the FJSP with maintenance activities.

3.1. Filtered beam search

Filtered beam search is an extension of beam search. Beam search is an adaptation of the branch-and-bound method in which it only explores the promising nodes level by level without backtracking. The number of nodes explored at each level is called the *beamwidth*. The node evaluation process at each level is a key issue in the beam search technique (Valente & Alves, 2005). An evaluation function is usually used to determine which nodes to continue the search on, which poses the problem of finding a good trade-off between quick, but poor evaluation and more computationally demanding, but better evaluation (Sabuncuoglu & Bayiz, 1999).

To find a good trade-off mentioned above economically and quickly, filtered beam search is introduced (Ow & Morton, 1988). Filtered beam search procedure uses both crude and accurate evaluations in two phases: filtering phase and beam selection phase. All nodes generated from a parent node are evaluated crudely by a computationally inexpensive filtering procedure (normally named as local evaluation function) in the first phase, leaving *filterwidth* filtered nodes for further accurate evaluation by a total cost evaluation function (normally named as global evaluation function) in the second phase. Those nodes found most promising are added to a partial solution. This procedure is repeated on parallel *beamwidth* paths until the complete schedules are reached. Hence, the number of solutions saved is equal to size of the *beamwidth*. Eventually, the solution with the best objective function value is as the final schedule.

As shown in Fig. 1, the promising nodes (beam nodes) are determined by applying local and global evaluation functions and proceed with the search through these selected nodes. After determining the beam nodes in level 1, the filtered beam search

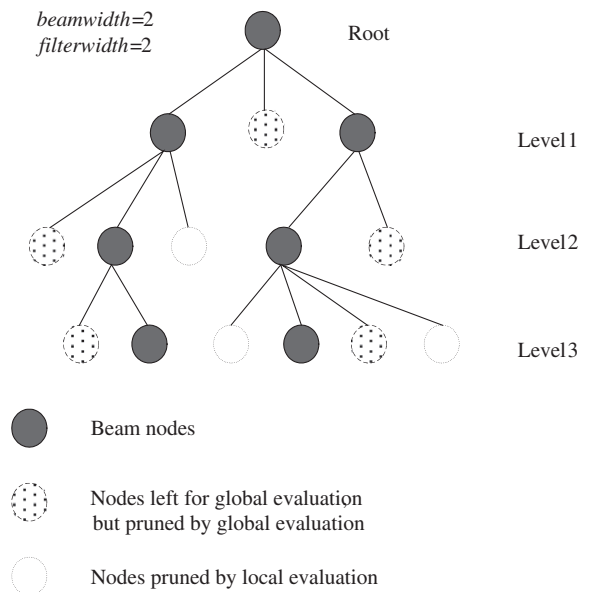


Fig. 1. Representation of filtered beam search tree.

procedure is employed independently to generate one partial tree from each of them (in Fig. 1, there are two partial trees since *beamwidth* = 2). These partial trees are referred as beam paths. In each beam path, once a beam node is determined, nodes are generated in next level from this beam node by applying branching scheme. These generated nodes first subject to filtering phase, in which some of them are pruned by local evaluation and *filterwidth* (*filterwidth* = 2 in Fig. 1) filtered nodes are left for beam selection phase. Through global evaluation function, one of the filtered nodes with best global evaluation function value is determined as beam node for the level. The procedure is repeated in Fig. 1 and at last it forms two beam paths.

3.2. Modified FBS-based heuristic for the FJSP with maintenance tasks

The proposed heuristic is the follow up of our previous work of Wang et al. (2008). As the FBS-based heuristic algorithm for FJSP, the modified FBS-based heuristic should also solve four major elements: (1) search tree representation to define a solution space; (2) determination of *beamwidth* and *filterwidth*; (3) branching scheme; (4) local and global evaluation functions selection. Among them, (1) and (2) are defined as same as those of Wang et al. (2008).

Branching scheme is modified to incorporate with the machine availability constraint and maintenance resource constraints.

Maintenance activities can be regarded as specific production jobs. In the case of sufficient maintenance resources, each maintenance activity can be seen as a job with only one operation without any precedent constraints. While in the case of only one maintenance resource, there is a precedent constraint among maintenance activities. To adapt to the maintenance activities in the FJSP, the branching scheme is a modification form of the M_NON-DELAY proposed in Wang et al. (2008). For differentiation, we call it MNew_NON here.

The basic idea of the MNew_NON is: maintenance activities are also considered as operations that should be processed only on the pre-determined machine with the earliest starting time S_u^k and latest starting time $\bar{S}_u^k$ ($S_u^k = \bar{S}_u^k$ for the fixed availability constraint). The bottom line is that the period from the latest starting time plus the duration for the maintenance task pm_u^k should be at least reserved for the maintenance task. It is possible that the maintenance task could be processed earlier if the system overall

criteria suggests based on the computation of the proposed FBS-based heuristic.

Firstly, to describe the MNew_NON, some notations are defined.

- PS_l is a partial schedule containing l scheduled operations (including maintenance activities)
- S_l is the set of schedulable operations (including maintenance activities) at level l corresponding to a given PS_l
- S_{IM} is the set of maintenance activities that are neither scheduled in PS_l nor in S_l
- M_{ij} is the set of machines can perform the operation O_{ij}
- T_{ijk} is the earliest time for an alternative machine that can be ready to perform the operation $O_{ij}, M_k \in M_{ij}$
- T_k is the earliest time for a machine on which the maintenance activity can be pre-scheduled (sometimes $T_{ijk} = T_k$)
- s_{ij} is the earliest time at which the operation $O_{ij} \in S_l$ could be started, and $s_{(i+1)j} = s_{ij} + \gamma_{ij}$, where $\gamma_{ij} = \min_k(p_{ijk})$
 $\forall 1 \leq i \leq n_j, 1 \leq j \leq n$ (for maintenance activities, $s_{ij} = \bar{S}_u^k$, $\gamma_{ij} = pm_u^k$, and if the maintenance activities are not scheduled in this level, its earliest time in the next level will be updated via $\max(\bar{S}_u^k, \min(T_k, \bar{S}_u^k))$. $s_{1j} = r_j$ (r_j is release time))
- O is a general operation, either an operation O_{ij} or a maintenance activity PM_u^k

Then the MNew_NON procedure is explained as follows:

- Step 1:** Determine $T^* = \min_{O \in S_l} \{s_{ij}\}$, and let the operations $O \in S_l$ with $s_{ij} = T^*$ be a set $\bar{S}_l$.
- Step 2:** Select an operation $O \in \bar{S}_l$, for each machine of M_{ij} , generate a new node that corresponds to the partial schedule, in which operation O is added to PS_l and started at time $s_{ijk} = \max(T^*, T_{ijk})$ (as in Fig. 2a, the maintenance activity is scheduled as an operation).
 If the operation is a maintenance activity and there are sufficient maintenance resources available, then update T_k and s_{ij} , and step out of the MNew_NON procedure.
 If the operation is a regular production operation, then go to Step 3.
- If the operation is a maintenance activity and there is only one maintenance resource available, then go to Step 4.
- Step 3:** If $s_{ijk} + p_{ijk} > \bar{S}_u^k$ (here, s_{ijk} must be less than or equal to $\bar{S}_u^k$ according to the design of this branching scheme) and $PM_u^k \in S_{IM}$ (that means there is no any room for the process of the maintenance activity on the same machine), the node $i-j-k$ (denotes that the operation O_{ij} is to be assigned on the machine M_k) will be replaced by the maintenance

task PM_u^k with the starting time $s_{ijk} = T_k$ (as shown in Fig. 2b). Meanwhile, the PM_u^k will not be included in $S_{(l+1)M}$. In the case of only one maintenance resource, go to Step 4; Otherwise, update T_{ijk} and s_{ij} , and step out of the MNew_NON procedure.

- Step 4:** For maintenance tasks in S_{IM} (PM_u^k scheduled is not included), their earliest starting time is determined by $\min(\max(T_k, s_{ijk} + pm_u^k), \bar{S}_u^k)$, $k \in \{1, 2, \dots, m\}$. Step out of the MNew_NON procedure. This is used to fix the sequence between maintenance activities for the case of only one maintenance resource available.

The MNew_NON procedure mentioned above is easy to incorporate the case of fixed machine availability constraint (i.e., set $\bar{S}_u^k = \bar{S}_u^k$).

As for local and global evaluation functions selection, the M_EET (modified earliest ending time) is used as both local and global evaluation functions. The M_EET is defined as: for all nodes generated using the MNew_NON procedure, favor the node with the earliest ending time (i.e., favor the node with $\min(s_{ijk} + p_{ijk})$ for regular job operations or $\min(s_{ijk} + pm_u^k)$ for maintenance activities). Balance the loads on machines for the tie breaking. Then, ties are broken by a random choice.

Based on the elements mentioned above, the main steps of the algorithm procedure are as follows.

FBS algorithm with MNew_NON procedure for FJSP with maintenance activities

Input: beamwidth b , filterwidth f , total number of general operations $N_0 = n + \sum_{k=1}^m L_k$, data set of scheduling problem with maintenance activities.

Output: the best schedule results

Begin

Step 1. Initialization

- $l \leftarrow 1$: search tree level;
- $bn \leftarrow 1$: beamwidth indicator;
- $PS_l \leftarrow \Phi$.

Step 2. Determining Initial Beam Nodes

(1) Generate nodes from the root node by using the procedure MNew_NON. Check the total number of generated nodes N . Update the PS_l with generated nodes.

(2) While ($N < b$),

Do{ $l \leftarrow l + 1$, generate new nodes (update N correspondingly) using the procedure MNew_NON with PS_l as the partial schedule, and update PS_l . }

(3) Compute the global evaluation function values using M_EET for all the nodes and select the best b number of nodes as initial beam nodes. Meanwhile, determine potential sets $PS_l(1), \dots, PS_l(b)$.

Step 3. Outer Loop: Determine b Beam Paths While ($bn < b$)

Do{ Inner Loop: Determine Beam Nodes of Each Level for Each Initial Beam Node While ($l < N_0$)

Do{

- (a) Generate new nodes (the number of nodes generated is N) from the beam node by using the procedure MNew_NON with $PS_l(bn)$ as the partial schedule. Using M_EET, compute the local evaluation function values for each node and select $f = \min(N, f)$ number of nodes with the best values for further evaluation.

- (b) Using M_EET, compute the global evaluation function values of f number of nodes, select the node with the best value and add the node into the partial schedule $PS_l(bn)$, update the $PS_l(bn)$.

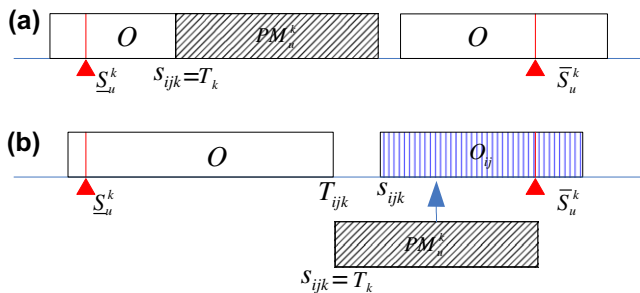


Fig. 2. A schematic illustration of the starting time of a maintenance activity in the branching scheme.

Table 1The PM activities in the problem $8 \times 8/27$ operations.

		PM ₁ ¹	PM ₁ ²	PM ₁ ³	PM ₁ ⁴	PM ₁ ⁵	PM ₁ ⁶	PM ₁ ⁷	PM ₁ ⁸
Time window	S_u^k	1	3	5	6	0	3	1	3
	$\bar{S}_u^k$	6	6	10	14	7	11	11	9
Duration	pm_{ij}^k	4	3	5	3	3	5	3	4

(c) $l \leftarrow l + 1$

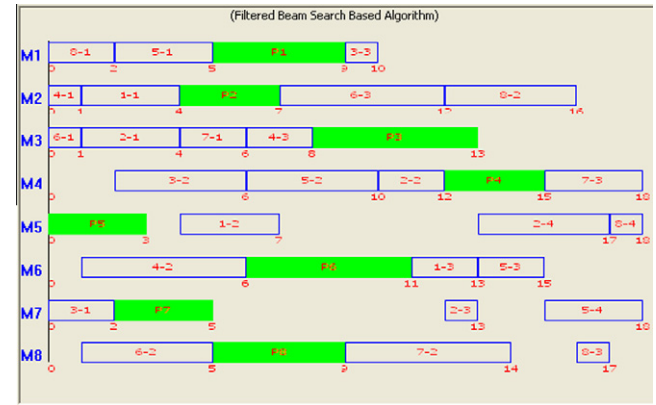
}
Formulate the **bn**-th complete schedule
 $PS(bn).bn \leftarrow bn + 1$
}

Step 4. Select the Solution Schedule. Select the schedule set or schedule sets with the best objective function values among the final b schedule sets and output.
End

4. Numerical results

4.1. Two general FJSP with maintenance activities

Two representative flexible job-shop scheduling problem benchmarking instances (denoted by problem $n \times m/N$ operations) based on practical data have been selected, which are from Kacem et al. (2002a), Kacem et al. (2002b), respectively. The considered FJSP with maintenance activities in this paper are extended from these two original FJSP problems by considering exactly one PM activity on each machine in the planning horizon. The results are obtained under an objective function $0.4 \times F_1 + 0.3 \times F_2 + 0.3 \times F_3$ with *beamwidth* $b = 5$ and *filterwidth* $f = 5$. As mentioned above, the combination (M_EET, M_EET) is used as local and global evaluation functions.

**Fig. 4.** The scheduling results of the problem $8 \times 8/27$ operations with maintenance activities with non-fixed starting time.**Table 2**The comparison of results with those of Gao et al. (2006) on the problem $8 \times 8/27$ operations.

	Gao et al. (2006)	FBS-based heuristic
F_1	17	18
F_2	105	103
F_3	15	16

4.1.1. Problem $8 \times 8/27$ operations with maintenance activities

4.1.1.1. Non-fixed availability constraint and sufficient maintenance resources. The non-fixed availability constraint is set as the same as in Gao et al. (2006) based on the original problem $8 \times 8/27$ operations (Kacem et al., 2002a; Wang et al., 2008). The time window of starting time and duration of maintenance tasks are shown in Table 1. For instance, there is a maintenance task needed to be

Jobno	jobind	MachSet	processtime	Jobnext	Jobpre	starttime	form_start	endtime	priority	due_date	Jobindex
1-1	0	1/2/3/4/5/6/7/8	5/3/5/3/3/9999/10/9	1-2	NULL	0	0	0	1		0-1-1
1-2	1	1/2/3/4/5/6/7/8	10/9999/5/8/3/9/9/6	1-3	1-1	3	0	0	1		0-1-2
1-3	2	1/2/3/4/5/6/7/8	9999/10/9999/5/6/2/4/5	NULL	1-2	6	0	0	1		0-1-3
2-1	3	1/2/3/4/5/6/7/8	5/7/3/9/8/9999/9/9999	2-2	NULL	0	0	0	1		0-2-1
2-2	4	1/2/3/4/5/6/7/8	9999/8/5/2/6/7/10/9	2-3	2-1	3	0	0	1		0-2-2
2-3	5	1/2/3/4/5/6/7/8	9999/10/9999/5/6/4/1/7	2-4	2-2	5	0	0	1		0-2-3
2-4	6	1/2/3/4/5/6/7/8	10/8/9/6/4/7/9999/9999	NULL	2-3	6	0	0	1		0-2-4
3-1	7	1/2/3/4/5/6/7/8	10/9999/9999/7/6/5/2/4	3-2	NULL	0	0	0	1		0-3-1
3-2	8	1/2/3/4/5/6/7/8	9999/10/6/4/8/9/10/9999	3-3	3-1	2	0	0	1		0-3-2
3-3	9	1/2/3/4/5/6/7/8	1/4/5/6/9999/10/9999/7	NULL	3-2	6	0	0	1		0-3-3
4-1	10	1/2/3/4/5/6/7/8	3/1/6/5/9/7/8/4	4-2	NULL	0	0	0	1		0-4-1
4-2	11	1/2/3/4/5/6/7/8	12/11/7/8/10/5/6/9	4-3	4-1	1	0	0	1		0-4-2
4-3	12	1/2/3/4/5/6/7/8	4/6/2/10/3/9/5/7	NULL	4-2	6	0	0	1		0-4-3
5-1	13	1/2/3/4/5/6/7/8	3/6/7/8/9/9999/10/9999	5-2	NULL	0	0	0	1		0-5-1
5-2	14	1/2/3/4/5/6/7/8	10/9999/7/4/9/8/6/9999	5-3	5-1	3	0	0	1		0-5-2
5-3	15	1/2/3/4/5/6/7/8	9999/9/8/7/4/2/7/9999	5-4	5-2	7	0	0	1		0-5-3
5-4	16	1/2/3/4/5/6/7/8	11/9/9999/6/7/5/3/6	NULL	5-3	9	0	0	1		0-5-4
6-1	17	1/2/3/4/5/6/7/8	6/7/11/4/6/9/9999/10	6-2	NULL	0	0	0	1		0-6-1
6-2	18	1/2/3/4/5/6/7/8	11/9999/9/9/9/7/6/4	6-3	6-1	1	0	0	1		0-6-2
6-3	19	1/2/3/4/5/6/7/8	10/5/9/10/11/9999/10/9999	NULL	6-2	5	0	0	1		0-6-3
7-1	20	1/2/3/4/5/6/7/8	5/4/2/6/7/9999/10/9999	7-2	NULL	0	0	0	1		0-7-1
7-2	21	1/2/3/4/5/6/7/8	9999/9/9999/9/11/9/10/5	7-3	7-1	2	0	0	1		0-7-2
7-3	22	1/2/3/4/5/6/7/8	9999/6/9/3/8/6/9999/10	NULL	7-2	7	0	0	1		0-7-3
8-1	23	1/2/3/4/5/6/7/8	2/8/5/9/9999/4/9999/10	8-2	NULL	0	0	0	1		0-8-1
8-2	24	1/2/3/4/5/6/7/8	7/4/7/8/9/9999/10/9999	8-3	8-1	2	0	0	1		0-8-2
8-3	25	1/2/3/4/5/6/7/8	9/9999/8/5/6/7/11	8-4	8-2	6	0	0	1		0-8-3
8-4	26	1/2/3/4/5/6/7/8	9/9999/3/7/11/5/8/9999	NULL	8-3	7	0	0	1		0-8-4
p-1	27	1	4	NULL	NULL	1	6	0	1		0P1
p-2	28	2	3	NULL	NULL	3	6	0	1		0P2
p-3	29	3	5	NULL	NULL	5	10	0	1		0P3
p-4	30	4	3	NULL	NULL	6	14	0	1		0P4
p-5	31	5	3	NULL	NULL	0	7	0	1		0P5
p-6	32	6	5	NULL	NULL	3	11	0	1		0P6
p-7	33	7	3	NULL	NULL	1	11	0	1		0P7
p-8	34	8	4	NULL	NULL	3	9	0	1		0P8

Fig. 3. The input data for the problem $8 \times 8/27$ operations with maintenance activities.

Table 3
The fixed starting time PM activities of the problem $8 \times 8/27$ operations.

		PM_1^1	PM_1^2	PM_1^3	PM_1^4	PM_1^5	PM_1^6	PM_1^7	PM_1^8
(a) Fixed as the earliest starting time of the time window									
Starting time	$\underline{S}_u^k = \bar{S}_u^k$	1	3	5	6	0	3	1	3
Duration	pm_u^k	4	3	5	3	3	5	3	4
(b) Fixed as the latest starting time of the time window									
Starting time	$\underline{S}_u^k = \bar{S}_u^k$	6	6	10	14	7	11	11	9
Duration	pm_u^k	4	3	5	3	3	5	3	4

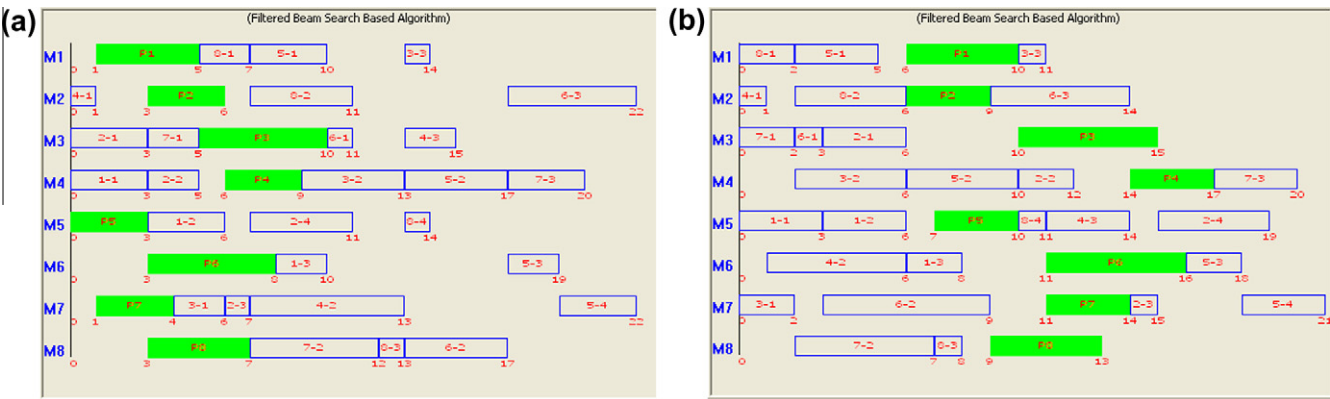


Fig. 5. The scheduling results of problem $8 \times 8/27$ operations with maintenance activities with fixed starting time. (a) The starting time of each maintenance task is set as the earliest starting time of the time window and (b) the starting time of each maintenance task is set as the latest starting time of the time window.

Table 4
The non-fixed starting time PM activities of the problem $8 \times 8/27$ operations in the case of only one maintenance resource.

		PM_1^1	PM_1^2	PM_1^3	PM_1^4	PM_1^5	PM_1^6	PM_1^7	PM_1^8
Time window	$\underline{S}_u^k$	3	3	5	6	0	3	1	1
	$\bar{S}_u^k$	10	8	10	19	11	18	14	15
Duration	pm_u^k	2	3	2	3	1	2	3	2

processed on machine 1. The earliest starting time is at one and the latest starting time is allowed to be at six and the duration of the maintenance task is four time units.

The input data of this problem is shown in Fig. 3, in which the “Jobno” with “p” stands for the maintenance tasks while others stands for the operations of regular jobs. The “MachSet” represents

the machine set in which any machine can process the corresponding operation, with the corresponding “processtime” labeled by “/”. “9999” in “processtime” indicates that the assignment is impossible. The precedence constraints are shown with “Jobnext” and “Jobpre”. “starttime” shows the earliest starting time, “form_start” shows the latest starting time for maintenance tasks (regular jobs are not considered). “Priority” and “due-date” are left for future extension for different system performance criteria associated with due date or priority weight. “JobIndex” is for labeling of scheduling output.

The scheduling results are shown in Fig. 4. The results are compared with those of Gao et al. (2006) (as shown in Table 2.). They employed a hybrid GA with pre-determined nine parameters. Though values of three different objectives obtained in the proposed method are not all better than those of Gao et al. (2006), F_2 (Total workload of the machines) obtained is better and two other objectives are closest to those of Gao et al. (2006) (only less

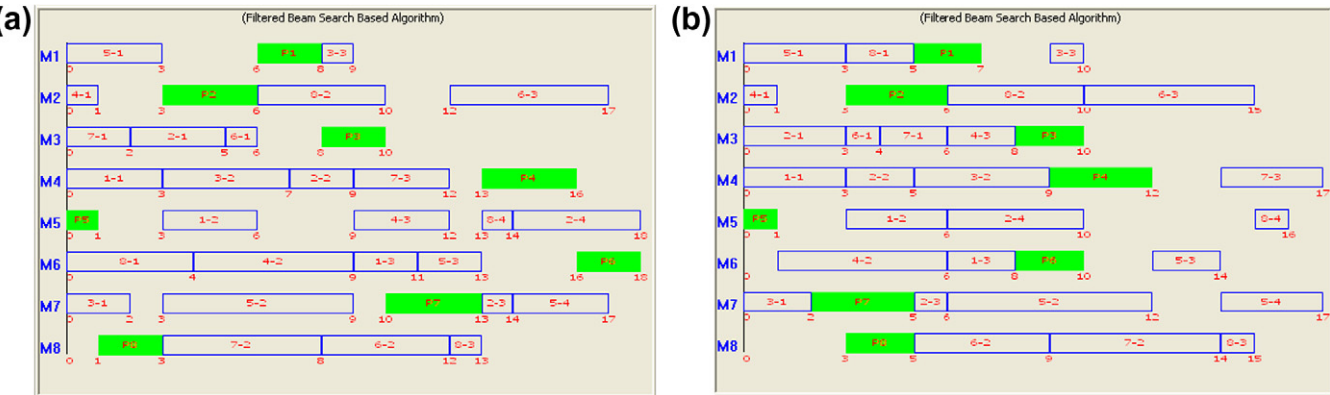
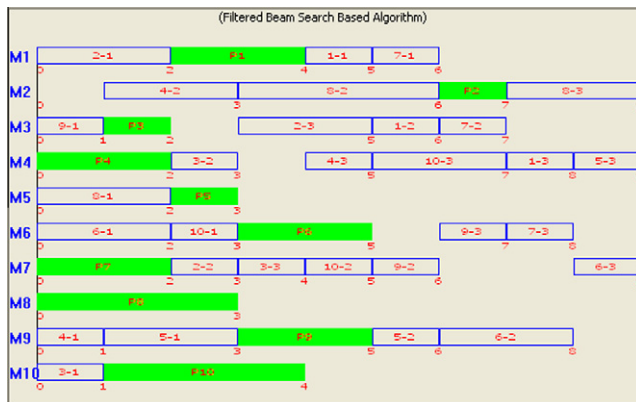


Fig. 6. The scheduling results of the problem $8 \times 8/27$ operations with maintenance activities in the case of (a) only one maintenance resource available and (b) the case of sufficient maintenance.

Table 5The PM activities in the problem $10 \times 10/30$ operations.

		PM ₁ ¹	PM ₁ ²	PM ₁ ³	PM ₁ ⁴	PM ₁ ⁵	PM ₁ ⁶	PM ₁ ⁷	PM ₁ ⁸	PM ₁ ⁹	PM ₁ ¹⁰
Time window	$\underline{S}_u^k$	0	1	0	0	2	0	0	0	0	0
	$\overline{S}_u^k$	2	6	5	3	6	4	3	4	3	3
Duration	pm_u^k	2	1	1	2	1	2	2	3	2	3

Jobno	jobind	MachSet	processtime	Jobnext	Jobpre	starttime	form_start	endtime	priority	due_date	JobIndex
1-1	0	1/2/3/4/5/6/7/8/9/10	1/4/6/9/3/5/2/8/9/5	1-2	NULL	0	0	0	1	0.1-1	
1-2	1	1/2/3/4/5/6/7/8/9/10	4/1/1/3/4/8/10/4/11/4	1-3	1-1	1	0	0	1	0.1-2	
1-3	2	1/2/3/4/5/6/7/8/9/10	3/2/5/1/5/6/9/5/10/3	NULL	1-2	2	0	0	1	0.1-3	
2-1	3	1/2/3/4/5/6/7/8/9/10	2/10/4/5/9/8/4/15/8/4	2-2	NULL	0	0	0	1	0.2-1	
2-2	4	1/2/3/4/5/6/7/8/9/10	4/8/7/1/9/6/1/10/7/1	2-3	2-1	2	0	0	1	0.2-2	
2-3	5	1/2/3/4/5/6/7/8/9/10	6/1/1/2/7/5/3/5/14/9/2	NULL	2-2	3	0	0	1	0.2-3	
3-1	6	1/2/3/4/5/6/7/8/9/10	8/5/8/9/4/3/5/3/8/1	3-2	NULL	0	0	0	1	0.3-1	
3-2	7	1/2/3/4/5/6/7/8/9/10	9/3/6/1/2/6/4/1/7/2	3-3	3-1	1	0	0	1	0.3-2	
3-3	8	1/2/3/4/5/6/7/8/9/10	7/1/8/5/4/9/1/2/3/4	NULL	3-2	2	0	0	1	0.3-3	
4-1	9	1/2/3/4/5/6/7/8/9/10	5/10/6/4/9/5/1/7/1/6	4-2	NULL	0	0	0	1	0.4-1	
4-2	10	1/2/3/4/5/6/7/8/9/10	4/2/3/8/7/4/6/9/8/4	4-3	4-1	1	0	0	1	0.4-2	
4-3	11	1/2/3/4/5/6/7/8/9/10	7/3/12/1/6/5/8/3/5/2	NULL	4-2	4	0	0	1	0.4-3	
5-1	12	1/2/3/4/5/6/7/8/9/10	7/10/4/5/6/3/5/15/2/6	5-2	NULL	0	0	0	1	0.5-1	
5-2	13	1/2/3/4/5/6/7/8/9/10	5/6/3/9/8/2/8/6/1/7	5-3	5-1	2	0	0	1	0.5-2	
5-3	14	1/2/3/4/5/6/7/8/9/10	6/1/4/1/10/4/3/11/13/9	NULL	5-2	3	0	0	1	0.5-3	
6-1	15	1/2/3/4/5/6/7/8/9/10	8/9/10/8/4/2/7/8/3/10	6-2	NULL	0	0	0	1	0.6-1	
6-2	16	1/2/3/4/5/6/7/8/9/10	7/3/12/5/4/3/6/9/2/15	6-3	6-1	2	0	0	1	0.6-2	
6-3	17	1/2/3/4/5/6/7/8/9/10	4/7/3/6/3/4/1/5/1/11	NULL	6-2	4	0	0	1	0.6-3	
7-1	18	1/2/3/4/5/6/7/8/9/10	1/7/8/3/4/9/4/13/10/7	7-2	NULL	0	0	0	1	0.7-1	
7-2	19	1/2/3/4/5/6/7/8/9/10	3/8/1/2/3/6/1/1/2/13/3	7-3	7-1	1	0	0	1	0.7-2	
7-3	20	1/2/3/4/5/6/7/8/9/10	5/4/2/1/2/1/8/14/5/7	NULL	7-2	2	0	0	1	0.7-3	
8-1	21	1/2/3/4/5/6/7/8/9/10	5/7/1/1/3/2/9/8/5/12/8	8-2	NULL	0	0	0	1	0.8-1	
8-2	22	1/2/3/4/5/6/7/8/9/10	8/3/10/7/5/1/3/4/6/8/4	8-3	8-1	2	0	0	1	0.8-2	
8-3	23	1/2/3/4/5/6/7/8/9/10	6/2/1/3/5/4/3/5/7/9/5	NULL	8-2	5	0	0	1	0.8-3	
9-1	24	1/2/3/4/5/6/7/8/9/10	3/9/1/3/8/1/6/7/5/4	9-2	NULL	0	0	0	1	0.9-1	
9-2	25	1/2/3/4/5/6/7/8/9/10	4/6/2/5/7/3/1/9/6/7	9-3	9-1	1	0	0	1	0.9-2	
9-3	26	1/2/3/4/5/6/7/8/9/10	8/5/4/8/6/1/2/3/10/12	NULL	9-2	2	0	0	1	0.9-3	
a-1	27	1/2/3/4/5/6/7/8/9/10	4/3/1/6/7/1/12/6/20/6	a-2	NULL	0	0	0	1	0.10-1	
a-2	28	1/2/3/4/5/6/7/8/9/10	3/1/8/1/9/4/1/4/17/15	a-3	a-1	1	0	0	1	0.10-2	
a-3	29	1/2/3/4/5/6/7/8/9/10	9/2/4/2/3/5/2/4/10/23	NULL	a-2	2	0	0	1	0.10-3	
P-1	30	1	2	NULL	NULL	0	2	0	1	0P1	
P-2	31	2	1	NULL	NULL	1	6	0	1	0P2	
P-3	32	3	1	NULL	NULL	0	5	0	1	0P3	
P-4	33	4	2	NULL	NULL	0	3	0	1	0P4	
P-5	34	5	1	NULL	NULL	2	6	0	1	0P5	
P-6	35	6	2	NULL	NULL	0	4	0	1	0P6	
P-7	36	7	2	NULL	NULL	0	3	0	1	0P7	
P-8	37	8	3	NULL	NULL	0	4	0	1	0P8	
P-9	38	9	2	NULL	NULL	0	3	0	1	0P9	
P-91	39	10	3	NULL	NULL	0	3	0	1	0P10	

Fig. 7. The input data for the problem $10 \times 10/30$ operations with maintenance activities.**Fig. 8.** The scheduling results of the problem $10 \times 10/30$ operations with maintenance activities with non-fixed starting time.

than one time unit for each). Therefore, the solution could be an alternative one for scheduling decision-makers with different preferences.

4.1.1.2. Fixed availability constraint and sufficient maintenance resources. To employ the proposed heuristic for this case, the only

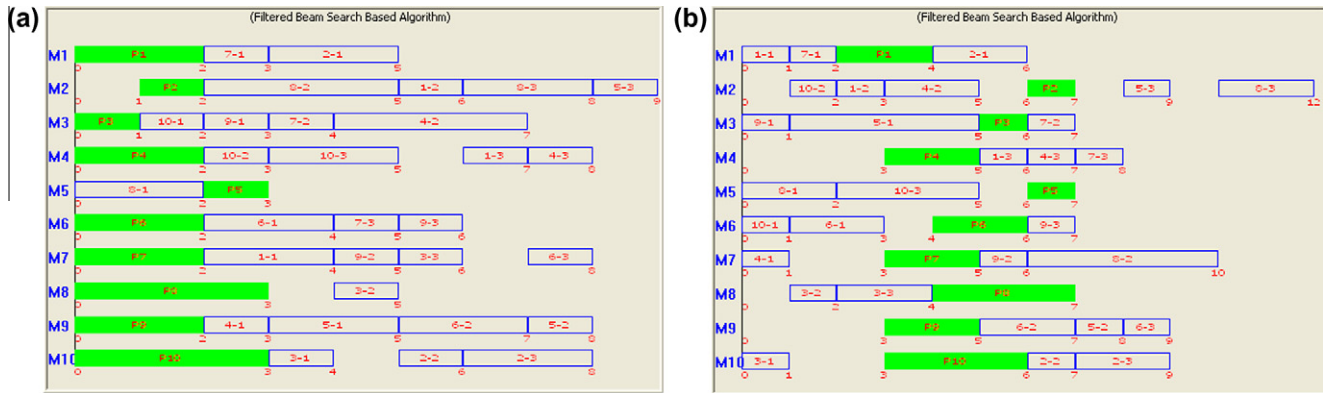
Table 6The comparison of results with those of Gao et al. (2006) on the problem $10 \times 10/30$ operations.

	Gao et al. (2006)	FBS-based heuristic
F_1	8	9
F_2	61	60
F_3	7	8

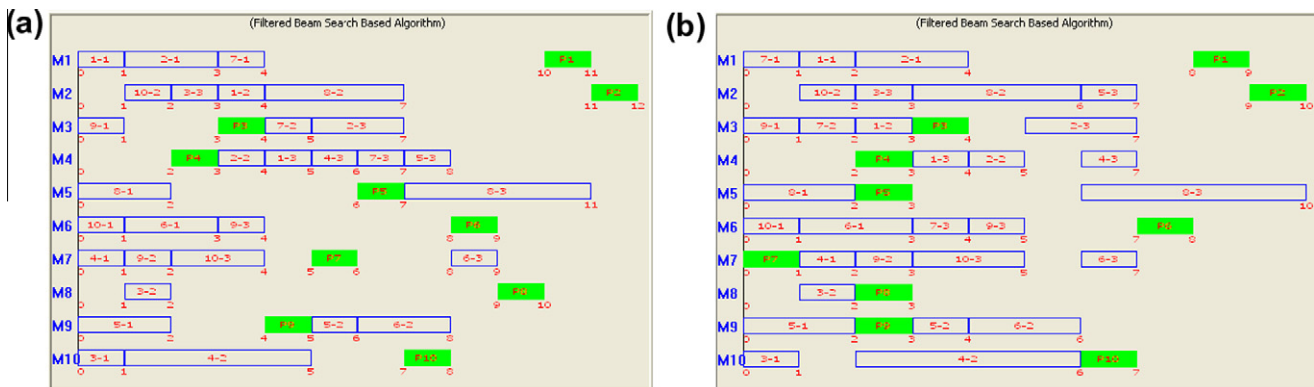
thing to do is to set values under “starttime” and “form_start” of maintenance activities in Fig. 3 as same. Assume that the starting time of each maintenance task is set as the earliest starting time (case (a)) and the latest starting time (case (b)) of the time window, separately, as shown in Table 3. That means, in case (a), all maintenance tasks are shifted left and to be processed as early as possible, and in case (b), all maintenance tasks are shifted right and to be processed as late as possible. The scheduling results are shown in Fig. 5, with $F_1 = 22$, $F_2 = 104$ and $F_3 = 19$ for case (a) and $F_1 = 21$, $F_2 = 106$ and $F_3 = 17$ for case (b). The results of both cases are worse than those of Fig. 4 ($F_1 = 18$, $F_2 = 103$ and $F_3 = 16$). This shows that under the same setting of algorithm parameters, the performance improvement of scheduling results benefits from jointly optimizing maintenance and scheduling.

Table 7The fixed starting time PM activities of the problem $10 \times 10/30$ operations.

		PM_1^1	PM_1^2	PM_1^3	PM_1^4	PM_1^5	PM_1^6	PM_1^7	PM_1^8	PM_1^9	PM_1^{10}
(a) Fixed as the earliest starting time of the time window											
Starting time	$S_u^k = \bar{S}_u^k$	0	1	0	0	2	0	0	0	0	0
Duration	pm_u^k	2	1	1	2	1	2	2	3	2	3
(b) Fixed as the latest starting time of the time window											
Starting time	$S_u^k = \bar{S}_u^k$	2	6	5	3	6	4	3	4	3	3
Duration	pm_u^k	2	1	1	2	1	2	2	3	2	3

**Fig. 9.** The scheduling results of problem $10 \times 10/30$ operations with maintenance activities with fixed starting time. (a) The starting time of each maintenance task is set as the earliest starting time of the time window and (b) The starting time of each maintenance task is set as the latest starting time of the time window.**Table 8**The non-fixed starting time PM activities of the problem $10 \times 10/30$ operations in the case of only one maintenance resource.

		PM_1^1	PM_1^2	PM_1^3	PM_1^4	PM_1^5	PM_1^6	PM_1^7	PM_1^8	PM_1^9	PM_1^{10}
Time window	$\underline{S}_u^k$	8	9	0	2	0	7	0	2	1	4
	$\bar{S}_u^k$	12	12	6	7	7	10	9	9	9	8
Duration	pm_u^k	1	1	1	1	1	1	1	1	1	1

**Fig. 10.** The scheduling results of the problem $10 \times 10/30$ operations with maintenance activities in the case of (a) only one maintenance resource available and (b) the case of sufficient maintenance resources.

4.1.1.3. Non-fixed availability constraint and only one maintenance resource. It is shown that in the cases (1) or (2), at least four maintenance resources are needed in order to execute the maintenance schedules simultaneously. Otherwise, the schedules are infeasible since there is no sufficient maintenance resource to process maintenance activities simultaneously.

In the work of Gao et al. (2006), sufficient maintenance resources are assumed implicitly. Here, the case of only one maintenance resource available with non-fixed machine availability constraint is considered. If only one maintenance resource are assumed in the case of fixed machine availability constraint, feasible solution is not always available, unless there is no overlap between

any two maintenance activities schedule on multiple machines. However, this cannot be ensured if the starting time is fixed. Therefore, only the case of non-fixed machine availability constraint and one maintenance resource are considered here.

To ensure that feasible solution exists, the characteristics of PM activities are designed as in Table 4, in which the latest starting times of PM activities are postponed to some extent and the durations of PM activities are relatively shortened. The schedule

results are shown in Fig. 6. Fig. 6a is the results subject to the constraint of only one maintenance resource available, with $F_1 = 18$, $F_2 = 96$ and $F_3 = 15$, while Fig. 6b is the results of sufficient maintenance resource constraint with $F_1 = 17$, $F_2 = 93$ and $F_3 = 15$. It shows that under the same setting of algorithm parameters, the scheduling results of sufficient maintenance resources are dominant over those of only one maintenance resource in this case.

Jobno	jobind	MachSet	processtime	Jobnext	Jobpre	starttime	form_start	endtime	priority	JobIndex
1-1	0	2	21	1-2	NULL	0	0	0	0.1	
1-2	1	3	34	1-3	1-1	21	0	0	0.1	
1-3	2	3/5	95/85	1-4	1-2	55	0	0	0.1	
1-4	3	1/2	53/63	1-5	1-3	150	0	0	0.1	
1-5	4	1/4	55/55	NULL	1-4	203	0	0	0.1	
2-1	5	4	52	2-2	NULL	0	0	0	0.2	
2-2	6	2/4/5	16/16/16	2-3	2-1	52	0	0	0.2	
2-3	7	1/2	71/71	2-4	2-2	68	0	0	0.2	
2-4	8	2/3/5	26/26/26	2-5	2-3	139	0	0	0.2	
2-5	9	1/5	21/21	NULL	2-4	165	0	0	0.2	
3-1	10	2/3	31/31	3-2	NULL	0	0	0	0.3	
3-2	11	1/5	12/12	3-3	3-1	31	0	0	0.3	
3-3	12	1/2	42/42	3-4	3-2	43	0	0	0.3	
3-4	13	4	39	3-5	3-3	85	0	0	0.3	
3-5	14	4/5	98/98	NULL	3-4	124	0	0	0.3	
4-1	15	4	77	4-2	NULL	0	0	0	0.4	
4-2	16	2	77	4-3	4-1	77	0	0	0.4	
4-3	17	1/4/5	79/79/79	4-4	4-2	154	0	0	0.4	
4-4	18	1/4	55/55	4-5	4-3	233	0	0	0.4	
4-5	19	3/4/5	86/86/86	NULL	4-4	288	0	0	0.4	
5-1	20	5	37	5-2	NULL	0	0	0	0.5	
5-2	21	3/4/5	34/34/34	5-3	5-1	37	0	0	0.5	
5-3	22	1/3	64/64	5-4	5-2	71	0	0	0.5	
5-4	23	2	19	5-5	5-3	135	0	0	0.5	
5-5	24	1	83	NULL	5-4	154	0	0	0.5	
6-1	25	2/3/5	43/43/43	6-2	NULL	0	0	0	0.6	
6-2	26	2/4/5	54/54/54	6-3	6-1	43	0	0	0.6	
6-3	27	1/2/5	92/92/92	6-4	6-2	97	0	0	0.6	
6-4	28	4	62	6-5	6-3	189	0	0	0.6	
6-5	29	2/3/5	79/79/79	NULL	6-4	251	0	0	0.6	
7-1	30	1/3	93/93	7-2	NULL	0	0	0	0.7	
7-2	31	4	69	7-3	7-1	93	0	0	0.7	
7-3	32	2/3/4	87/87/87	7-4	7-2	162	0	0	0.7	
7-4	33	3/5	77/77	7-5	7-3	249	0	0	0.7	
7-5	34	3	87	NULL	7-4	326	0	0	0.7	
8-1	35	1	60	8-2	NULL	0	0	0	0.8	
8-2	36	1/2	41/41	8-3	8-1	60	0	0	0.8	
8-3	37	1/3	38/38	8-4	8-2	101	0	0	0.8	
8-4	38	2/5	83/83	8-5	8-3	139	0	0	0.8	
8-5	39	4	24	NULL	8-4	222	0	0	0.8	
9-1	40	3/4	98/98	9-2	NULL	0	0	0	0.9	
9-2	41	1/4	17/17	9-3	9-1	98	0	0	0.9	
9-3	42	5	25	9-4	9-2	115	0	0	0.9	
9-4	43	1	44	9-5	9-3	140	0	0	0.9	
9-5	44	2/4	49/49	NULL	9-4	184	0	0	0.9	
A-1	45	1/3	96/96	A-2	NULL	0	0	0	0.10	
A-2	46	1/5	77/77	A-3	A-1	96	0	0	0.10	
A-3	47	2/4	79/79	A-4	A-2	173	0	0	0.10	
A-4	48	1/2/4	75/75/75	A-5	A-3	252	0	0	0.10	
A-5	49	2/3/5	43/43/43	NULL	A-4	327	0	0	0.10	
B-1	50	5	28	B-2	NULL	0	0	0	0.11	
B-2	51	3	35	B-3	B-1	28	0	0	0.11	
B-3	52	1	95	B-4	B-2	63	0	0	0.11	
B-4	53	4/5	76/76	B-5	B-3	159	0	0	0.11	
B-5	54	2	7	NULL	B-4	234	0	0	0.11	
C-1	55	1/5	81/81	C-2	NULL	0	0	0	0.12	
C-2	56	3/5	10/10	C-3	C-1	61	0	0	0.12	
C-3	57	1/2/3	95/95/95	C-4	C-2	71	0	0	0.12	
C-4	58	2/3	9/9	C-5	C-3	166	0	0	0.12	
C-5	59	2/4	35/35	NULL	C-4	175	0	0	0.12	
D-1	60	2/4/5	59/59/59	D-2	NULL	0	0	0	0.13	
D-2	61	1/4	16/16	D-3	D-1	59	0	0	0.13	
D-3	62	2	91	D-4	D-2	75	0	0	0.13	
D-4	63	3	59	D-5	D-3	166	0	0	0.13	
D-5	64	1/3/5	46/46/46	NULL	D-4	225	0	0	0.13	
E-1	65	1/5	43/43	E-2	NULL	0	0	0	0.14	
E-2	66	1	52	E-3	E-1	43	0	0	0.14	
E-3	67	1	28	E-4	E-2	95	0	0	0.14	
E-4	68	1/3/5	27/27/27	E-5	E-3	123	0	0	0.14	
E-5	69	2/4	50/50	NULL	E-4	150	0	0	0.14	
F-1	70	1/3	87/87	F-2	NULL	0	0	0	0.15	
F-2	71	2	45	F-3	F-1	87	0	0	0.15	
F-3	72	3	39	F-4	F-2	132	0	0	0.15	
F-4	73	1/2/5	9/9/9	F-5	F-3	171	0	0	0.15	
F-5	74	4	41	NULL	F-4	180	0	0	0.15	
P-1	75	1	49	NULL	NULL	201	201	0	0.P11	
P-2	76	1	49	NULL	NULL	463	463	0	0.P12	
P-3	77	2	35	NULL	NULL	104	104	0	0.P21	
P-4	78	2	68	NULL	NULL	520	520	0	0.P22	
P-5	79	3	98	NULL	NULL	233	233	0	0.P31	
P-6	80	3	29	NULL	NULL	499	499	0	0.P32	
P-7	81	4	49	NULL	NULL	137	137	0	0.P41	
P-8	82	4	49	NULL	NULL	507	507	0	0.P42	
P-9	83	5	58	NULL	NULL	129	129	0	0.P51	
P-91	84	5	98	NULL	NULL	783	783	0	0.P52	

Fig. 11. The input data for the MPM job-shop scheduling problem with maintenance activities.

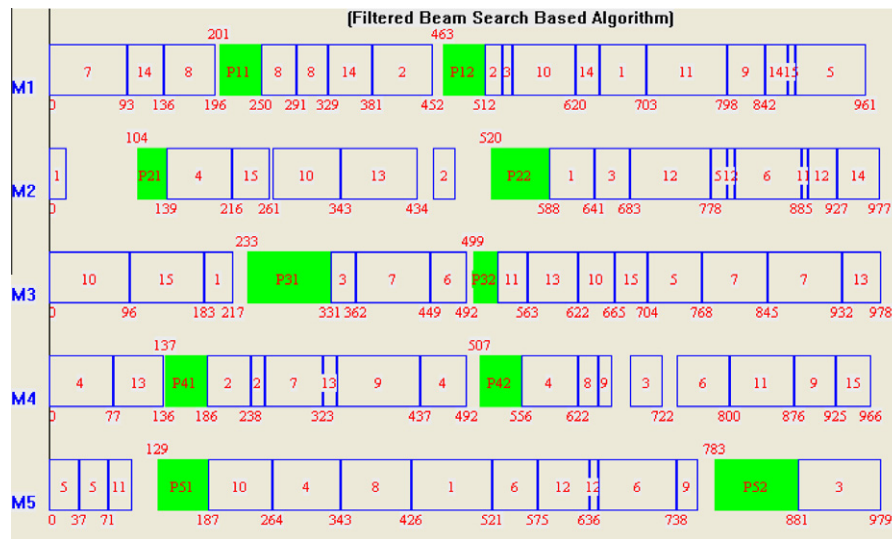


Fig. 12. The scheduling results of the MPM job-shop scheduling problem with maintenance activities.

4.1.2. Problem $10 \times 10/30$ operations with maintenance activities

4.1.2.1. Non-fixed availability constraint and sufficient maintenance resources. The non-fixed availability constraint is set as the same as in Gao et al. (2006) based on the original Problem $10 \times 10/30$ operations. The time window of starting time and duration of maintenance tasks are shown in Table 5. The input data of this problem is shown in Fig. 7. The scheduling results are shown in Fig. 8. The scheduling results are compared with those of Gao et al. (2006) (as shown in Table 6). Similar with the comparisons in Table 2, F_2 (Total workload of the machines) obtained is better and two other objectives are closest to those of Gao et al. (2006) (only less than one unit time for each). Therefore, the solution could be an alternative one to scheduling decision-makers with different preferences.

4.1.2.2. Fixed availability constraint and sufficient maintenance resources. Similar with the problem $8 \times 8/27$ operations with maintenance activities, assume that the starting time of each maintenance task is set as the earliest starting time (case (a)) and the latest starting time (case (b)) of the time window, separately, as shown in Table 7. The scheduling results are shown in Fig. 9, with $F_1 = 9$, $F_2 = 62$ and $F_3 = 8$ for case (a) and $F_1 = 12$, $F_2 = 65$ and $F_3 = 8$ for case (b). The results of both cases are worse than those of Fig. 8 ($F_1 = 9$, $F_2 = 60$ and $F_3 = 8$). This shows that under the same setting of algorithm parameters, the performance improvement of scheduling results benefits from jointly optimizing maintenance and scheduling.

4.1.2.3. Non-fixed availability constraint and only one maintenance resource. To ensure that feasible solution exists, the characteristics of PM activities are designed as in Table 8. The schedule results are shown in Fig. 10. Fig. 10a is the results subject to the constraint of only one maintenance resource available with $F_1 = 12$, $F_2 = 55$ and $F_3 = 7$, while Fig. 10b is the results of sufficient maintenance resource constraint with $F_1 = 10$, $F_2 = 55$ and $F_3 = 7$. It shows that under the same setting of algorithm parameters, the scheduling results with sufficient maintenance resources dominate those of only one maintenance resource if feasible solutions exist for the problem.

4.2. An example of MPM job-shop scheduling problem with maintenance activities

The MPM job-shop scheduling problem with maintenance activities is taken from Zribi et al. (2008). It is made of 15 jobs, five

operations per job and five machines. Each machine is subject to two maintenance periods as follows:

M_1 : [201,250], [463,512] means that the machine M_1 is unavailable between 201 and 250 time units and between 463 and 512 time units. M_2 : [104,139], [520,588], M_3 : [233,331], [499,528], M_4 : [137,186], [507,556], M_5 : [129,187], [783,881]. The objective is to minimize the makespan. Zribi et al. (2008) employed a two-stage heuristic: an assignment heuristic for the assignment problem, which then could be improved by a tabu search (TS), and a genetic algorithm for the sequencing problem. With the parameters setting of genetic algorithm (population size = 200, crossover rate = 0.8, mutation rate = 0.2, selection method = wheel, generations number = 400, number of execution = 10), the schedule result with assignment heuristic and GA is 1023, and the result with assignment heuristic enhanced by TS and GA is 968.

In this paper, the problem is computed with the proposed heuristic method. The input data of the problem is shown in Fig. 11. The beamwidth $b = 7$ and filterwidth $f = 7$. The obtained schedule results are shown in Fig. 12 with the makespan 979. The result is much better than that of the two-stage method of assignment heuristic and GA, and is very close to that of the combined method with assignment heuristic enhanced by TS and GA.

5. Conclusions

Production scheduling with machine availability constraints due to preventive maintenance has attracted more and more researchers, as the importance of applications is recognized. In this work, a scheduling problem with both fixed and non-fixed machine availability constraints due to maintenance activities is considered in the flexible job-shop manufacturing environment. Two cases of maintenance resources (i.e., unlimited maintenance resources and only one maintenance resource) are also considered. Thus, the extended flexible job-shop scheduling problem with maintenance activities is a much closer to the scheduling problems in realistic applications and possesses much more difficulties.

A heuristic based on filtered beam search (FBS) algorithm is proposed to solve this variant flexible job-shop scheduling problem (FJSP) with maintenance activities and maintenance resources constraint. A modified branching scheme is provided to easily incorporate both machine availability constraints and maintenance resource constraints. In order to test the effectiveness and easiness of the proposed heuristic, two representative experimental examples extended from the standard FJSP benchmarking problems and

a MPM job-shop scheduling problem with maintenance activities are tested. The results show that the proposed heuristic can obtain satisfactory solutions.

For the future work, the problem of maintenance resources optimality for the FJSP problem considering preventive maintenance with *nonresumable* jobs and *resumable* jobs in terms of manufacturing cost related performance measurement will be further investigated. Moreover, current work still belongs to the deterministic scheduling with machine availability constraints, in which maintenance activities are given as problem input data, not as a result of the machine operation (e.g., a function of machining time). Future work will also concern the jointly optimization problem considering PM policy selection, PM operations scheduling and production scheduling.

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